$$\mathbb{P} = \begin{array}{c} \\ A \\ D \\ W \end{array} \begin{array}{c} \begin{array}{ccc} A & D & W \end{array} \\ \begin{bmatrix} 0.05 & 0.65 & 0.30 \\ 0.35 & 0.40 & 0.25 \\ 0.30 & 0.55 & 0.15 \end{bmatrix} \end{array}$$

$$\mathbb{P}^2 = \begin{array}{c} \\ A \\ D \\ W \end{array} \begin{array}{c} \begin{array}{ccc} A & D & W \end{array} \\ \begin{bmatrix} 0.3200 & 0.4575 & 0.2225 \\ 0.2325 & 0.5250 & 0.2425 \\ 0.2525 & 0.4975 & 0.2500 \end{bmatrix} \end{array}$$

$$\mathbb{P}^3 = \begin{array}{c} \\ A \\ D \\ W \end{array} \begin{array}{c} \begin{array}{ccc} A & D & W \end{array} \\ \begin{bmatrix} 0.242875 & 0.513375 & 0.243750 \\ 0.268125 & 0.494500 & 0.237375 \\ 0.261750 & 0.500625 & 0.237625 \end{bmatrix} \end{array}$$

$$\mathbb{P}^4 = \begin{array}{c} \\ A \\ D \\ W \end{array} \begin{array}{c} \begin{array}{ccc} A & D & W \end{array} \\ \begin{bmatrix} 0.264950 & 0.497281 & 0.237769 \\ 0.257694 & 0.502638 & 0.239669 \\ 0.259594 & 0.501081 & 0.239325 \end{bmatrix} \end{array}$$

$$\mathbb{P}^5 = \begin{array}{c} \\ A \\ D \\ W \end{array} \begin{array}{c} \begin{array}{ccc} A & D & W \end{array} \\ \begin{bmatrix} 0.258627 & 0.501903 & 0.239471 \\ 0.260708 & 0.500374 & 0.238918 \\ 0.260156 & 0.500797 & 0.239047 \end{bmatrix} \end{array}$$

$$\mathbb{P}^{10} = \begin{array}{c} \\ A \\ D \\ W \end{array} \begin{array}{c} \begin{array}{ccc} A & D & W \end{array} \\ \begin{bmatrix} 0.260038 & 0.500871 & 0.239092 \\ 0.260034 & 0.500874 & 0.239093 \\ 0.260035 & 0.500873 & 0.239093 \end{bmatrix} \end{array}$$

$$\mathbb{P}^{10} = \begin{array}{c} \\ A \\ D \\ W \end{array} \begin{array}{c} \begin{array}{ccc} A & D & W \end{array} \\ \begin{bmatrix} 0.260038 & 0.500871 & 0.239092 \\ 0.260034 & 0.500874 & 0.239093 \\ 0.260035 & 0.500873 & 0.239093 \end{bmatrix} \end{array}$$

It appears that the long-run probability of finding the taxi at A, D, or W is

0.260, 0.501, and 0.239,

respectively.